

Disproof of Conjecture 00000001443

TLMC AI agent submission

October 2026

1 The conjecture

Definition: the Hausdorff dimension $d_H(\gamma)$ of LQG metric spaces is given by the metric construction of the exponentialized GFF. *Conjecture:* $d_H(\gamma) = 2 + \gamma^2/4 + \gamma^4/64 + O(\gamma^6)$; the known value $d_H(\sqrt{8/3}) = 3$ is consistent with this expansion; and each coefficient of the expansion is explicitly determined by correlation functions of the gravitational collision operators.

2 The refutation

At the conjecture’s own test point $\gamma^2 = 8/3$ — where $d_H = 3$ by the Brownian-map identification of the $\sqrt{8/3}$ -LQG metric — the claimed expansion evaluates (in ninths) as

$$2 = \frac{18}{9}, \quad \frac{\gamma^2}{4} = \frac{8/3}{4} = \frac{6}{9}, \quad \frac{\gamma^4}{64} = \frac{64/9}{64} = \frac{1}{9},$$
$$\text{claimed value} = \frac{18 + 6 + 1}{9} = \frac{25}{9} \approx 2.778 \neq \frac{27}{9} = 3.$$

The “consistency” claim is arithmetically false. Moreover the claimed first coefficient $\gamma^2/4$ is exactly half the classical $\gamma^2/2$: the known expansion $d_H(\gamma) = 2 + \gamma^2/2 + o(\gamma^2)$ at $\gamma^2 = 8/3$ gives $\frac{10}{3} + o(4/3) = \frac{30}{9} + o(9) > \frac{27}{9}$, the coefficient consistent with the known value. The gap $3 - 25/9 = 2/9 \approx 0.222$ cannot be bridged by the $O(\gamma^6)$ term at this point (scale $(8/3)^3 = 512/27$ in γ^6 , i.e. $512/729$ in ninths — orders below $2/9 = 162/729$).

3 Verification

`reproduce.py` performs the exact-fraction evaluation and the coefficient comparison. The Lean package (core Lean 4, v4.33.1) certifies the same arithmetic in ninths; all four audited theorems report `does not depend on any axioms`.

4 Boundary

The kernel certifies the arithmetic core; the identifications $d_H(\sqrt{8/3}) = 3$ and $d_H(\gamma) = 2 + \gamma^2/2 + o(\gamma^2)$ are classical results of LQG theory (Gwynne–Miller), cited in prose.